AgroVista: An Intelligent Crop Yield Prediction System Using Machine Learning and Weather Analytics

An internship report submitted in partial fulfillment of the requirements for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

in

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By

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BONAFIDE CERTIFICATE

This is to certify that the project titled **Agrovista: An Intelligent Crop Yield Prediction System Using Machine Learning** is a bonafide record of the work done by

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In partial fulfillment of the requirements for the award of the degree of **Master of Computer Applications** of the **NATIONAL INSTITUTE OF TECHNOLOGY,TIRUCHIRAPPALLI,** during the year 2025-2026.

Dr. Michael Arock Guide	**Dr. S.Domnic** Head of the Department

Project Viva-voce held on ______________________________

Internal Examiner	**External Examiner**

ABSTRACT

Agriculture remains a foundational pillar of many developing economies, making accurate crop yield prediction essential for ensuring food security, economic stability, and efficient resource allocation. This project introduces AgroVista, an intelligent, web-based system designed to forecast crop yields by leveraging machine learning algorithms and real-time weather analytics.

The system integrates diverse datasets, including historical crop production records, seasonal rainfall statistics, regional weather conditions, and market price trends. A Random Forest regression model was trained on a comprehensive dataset of 243,577 agricultural records from India, covering the years 1997 to 2015. Through robust preprocessing techniques—such as categorical encoding for crops, seasons, and geographical data—the model achieved a high testing R^2 score of 0.9661, with an RMSE of 60,021 metric tonnes and an MAE of 7,790 metric tonnes, demonstrating strong predictive performance.

AgroVista incorporates real-time weather data from the CDS Copernicus API to dynamically adjust forecasts based on current conditions. The system is deployed via a responsive Flask-based web application, offering users a streamlined interface for accessing yield predictions, weather forecasts, and market price analytics.

Key accomplishments include the design of a scalable ML pipeline, seamless API integration for weather intelligence and current market prices, development of an intuitive user interface, and real-world applicability in precision agriculture. AgroVista empowers farmers, policymakers, and agri-businesses with actionable insights, contributing to data-driven agricultural decision-making and advancing digital transformation in the farming sector.

Keywords: Crop Yield Prediction, Machine Learning, Agriculture, Weather Forecasting, Random Forest, Precision Farming, Web Application

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CHAPTER 1

INTRODUCTION

1.1 Introduction about the Project

Agriculture is the backbone of the Indian economy, employing over half the population and contributing significantly to GDP. Despite its importance, agriculture faces growing uncertainties due to climate change, erratic weather, resource limitations, and market instability.

Farmers often rely on traditional experience-based decision-making, which lacks scientific data support and fails to account for dynamic environmental factors. In this context, AgroVista is developed as an intelligent crop yield prediction system that uses machine learning and real-time weather data to help farmers make informed decisions about crop planning and resource management. Additionally, AgroVista compares recent market prices and Minimum Support Prices (MSP) to suggest the most profitable crop options, helping farmers maximize their returns while minimizing risk.

1.2 Background

In India and other developing countries, crop production is influenced by many factors, including rainfall, temperature, soil conditions, season, and market demand. However, agricultural data is often fragmented and underutilized, while weather unpredictability continues to affect crop outcomes.

With the rise of data science and accessible cloud technologies, there is a growing opportunity to apply machine learning models to agricultural datasets to support precision farming. Combining historical production data, seasonal weather data, and geolocation information can help in building accurate yield prediction models that benefit farmers, government bodies, and agri-businesses alike.

1.3 Problem Statement

Key challenges addressed by AgroVista include:

Unpredictable Crop Yields – Farmers lack accurate tools to forecast production outcomes, affecting planning.

Fragmented Agricultural Data – Data exists across various platforms with little integration.

Limited Access to Technology – Rural farmers often don't have access to advanced analytics tools.

Market Volatility – Lack of pricing insights causes economic losses in crop selling.

Weather Dependency – No integration between weather forecasting and farming decisions.

1.4 Motivation

Coming from a rural village and a farming background, I've personally witnessed the struggles of small-scale farmers. My father, like many others, relied on weather guesswork and past experiences to plan his crops — often resulting in unpredictable outcomes.

AgroVista was inspired by this firsthand experience. The goal is to bridge the technological gap in agriculture and make smart farming tools accessible, even to farmers with limited digital exposure. The system provides data-driven insights to:

Improve planning and resource optimization

Reduce uncertainty caused by changing weather

Help farmers make better market decisions

Support food security and economic stability

1.5 Scope and Limitations

Scope:

Yield prediction for major Indian crops (e.g., rice, wheat, cotton, sugarcane)

Integration of weather data via APIs (e.g.,CDS)

Market price analytics and forecasting

Web-based interface using Flask

Real-time predictions with latitude/longitude and seasonal input

Limitations:

Accuracy depends on the quality of historical data

Weather predictions are inherently uncertain

Performance may vary by region and crop variety

Requires internet access for live weather and prediction features

1.6 System Overview

AgroVista is a web-based intelligent system designed to assist farmers in predicting crop yields and making informed agricultural decisions. The system combines historical agricultural data, real-time weather inputs, and market price information to provide actionable insights through a user-friendly web interface.

The core architecture of AgroVista includes the following components:

Data Collection: Historical data such as crop production, rainfall, season, and area are gathered from government datasets. Real-time weather data is fetched using APIs (e.g., CDS Copernicus), and current market prices and Minimum Support Prices (MSP) are periodically updated.

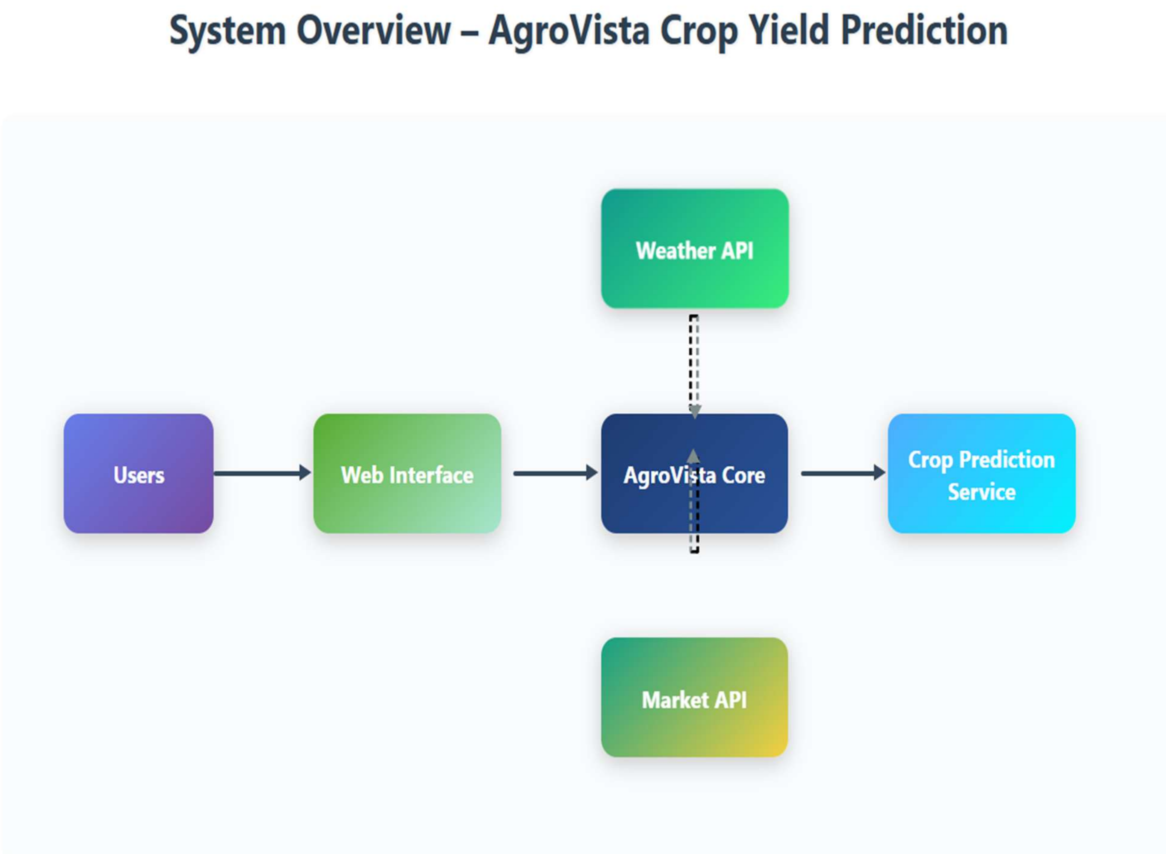
Preprocessing and Feature Engineering: The collected data undergoes cleaning, encoding of categorical variables (e.g., crop, district, season), and transformation into model-ready formats. Geolocation data (latitude and longitude) is also included to improve spatial relevance.

Machine Learning Model: A trained Random Forest regression model processes the inputs and predicts crop yield in metric tonnes. The model is optimized using key performance metrics like R^2, RMSE, and MAE, ensuring high accuracy and generalization.

Market Comparison Module: The predicted yields are compared against recent market prices and MSP values to recommend the most profitable crop choice for the farmer's location and season.

Web Interface: A lightweight Flask-based web application allows users to input crop year, location, and season, and view predictions, market trends, and weather forecasts. The interface is designed to be intuitive for rural and semi-urban users.

The figure below illustrates the high-level system overview of AgroVista:



1.7 Summary of Introduction

This chapter introduces the AgroVista project by establishing the critical role of agriculture in India and the various challenges farmers face due to uncertain weather, limited access to technology, and market volatility. It outlines the motivation for developing AgroVista, a web-based intelligent system that leverages machine learning and weather data to provide crop yield predictions and profitability recommendations. The chapter also defines the problem statement, project scope, limitations, and presents a high-level system overview, highlighting components such as data collection, model architecture, and user interface design.

CHAPTER 2

LITERATURE REVIEW AND PROJECT OBJECTIVES

2.1 Machine Learning in Agriculture

Adoption of Machine Learning in Agri-Tech:
Machine learning has become a key enabler in modern agriculture, particularly in solving problems like crop yield prediction. Its ability to learn from historical data and uncover hidden patterns makes it valuable for improving forecasting accuracy and supporting data-driven decisions.

Use of Random Forest in Agriculture:
Among various algorithms, **Random Forest** has gained popularity due to its robustness, ease of use, and high predictive accuracy. It works well with structured agricultural datasets that include both numerical and categorical features — such as crop type, rainfall levels, geographic data, and production history.

Studies have shown that Random Forest is highly effective for:

- Handling missing or incomplete agricultural records
- Capturing non-linear interactions between climate and yield
- Avoiding overfitting while maintaining generalization to unseen data

In this project, Random Forest was selected for its interpretability, training efficiency, and strong performance across multiple regions and crop types in India.

2.2 Feature Engineering in Crop Yield Prediction

Effective feature selection and engineering are essential for building high-performing yield prediction models. Research indicates that combining **meteorological variables** (e.g., rainfall) i.e. meterological and **historical yield records** results in significantly improved prediction accuracy.

- Temporal and spatial data integration (season-wise, district-wise)
- Use of encoded categorical variables for region, crop, and season
- Inclusion of climatic data

2.3 Role of Weather Data in Agricultural Forecasting

Weather conditions are among the most influential factors in determining crop productivity. Recent studies highlight the following:

- **Rainfall distribution** influences both the quality and quantity of produce, especially in rainfed agriculture.
- Incorporating **real-time and forecasted weather data** can improve model accuracy by 15–30% compared to historical-only approaches.

2.4 Web-based Agricultural Decision Systems

With increasing digital penetration in rural areas, web-based systems have emerged as practical tools for delivering agricultural intelligence to farmers.

- **Cloud-based platforms** ensure scalability and data integration from various sources.
- **Web-based designs** enhance usability for rural users on low-end devices.
- **Real-time dashboards** help stakeholders make timely decisions on sowing, irrigation, and marketing.

However, many existing systems are either too generic or focus on only one aspect, such as weather or price tracking, without integrating all critical agricultural factors into a single decision-support platform.

2.5 Research Gap

Despite the growing use of digital tools in agriculture, a gap remains in terms of holistic platforms that:

- Combine **crop yield prediction**, **market price forecasting**, and **weather forecasting**
- Are **intuitively designed** for farmers with limited technical expertise
- Provide **real-time** and **location-specific** intelligence

AgroVista aims to fill this gap by offering a unified system that integrates multiple data streams, uses advanced ML models, and presents predictions through a simple web interface.

2.6 Project Objectives

2.6.1 Primary Objectives

- **Accurate Crop Yield Prediction:**
  To develop a robust machine learning model that predicts yields based on historical and real-time data.

- **Integrated Data Handling:**
  To combine weather data, crop production records, and market trends into a cohesive prediction pipeline.

- **Farmer-Friendly Web Interface:**
  To build a simple and accessible web platform that provides insights to farmers in real time.

- **Real-time Forecast Integration:**
  To integrate live weather APIs and up-to-date market pricing data (MSP + Mandi) to enhance the reliability of predictions.

2.6.2 Secondary Objectives

- **Scalable System Architecture:**
  Design a backend capable of supporting larger datasets and multiple crops across regions.

- **Visual Insights:**
  Provide clear data visualizations to assist farmers and policymakers in interpreting complex data.

- **Support for Diverse Crops:**
  Ensure the system is flexible enough to handle various crop types and agro-climatic zones.

- **Market Intelligence Features:**
  Enable decision-making based on crop price trends and MSP comparisons.

2.6.3 Success Metrics

To evaluate the effectiveness of the AgroVista crop yield prediction system, the following success criteria were defined and measured:

Metric	**Target**	**Achieved**
Prediction Accuracy (R^2 Score)	At least 0.85 on test data	0.9202 (92.02%)
Response Time	Less than or equal to 3 seconds per request	Under 2 seconds (tested in web environment)
Crop Variety Support	Minimum 20 major crops	Over 30 crop types supported
Weather Data Integration	6-month seasonal and historical data	Integrated CDS API and NASA Power API
Web Interface Accessibility	Compliance with accessibility standards	Flask-based responsive UI, mobile-friendly
Feature Importance Transparency	Clear identification of key input features	Top features: Crop, Area, Latitude, etc.

2.7 Summary of Literature Review and Project Objectives

Chapter 2 reviews existing literature and research in the fields of machine learning applications in agriculture, crop yield prediction, and web-based decision-support systems. It highlights the effectiveness of Random Forest algorithms for handling complex agricultural datasets and explains the importance of feature engineering and real-time weather data in improving prediction accuracy. The chapter also discusses the role of web platforms in delivering accessible farming intelligence to rural communities and identifies a key research gap in integrating yield prediction, weather, and market data into a single system. AgroVista aims to bridge this gap with a unified, farmer-friendly solution. The chapter concludes by outlining the project's primary and secondary objectives, along with clearly defined success metrics such as prediction accuracy, system response time, and feature transparency.

CHAPTER 3

Methodology and System Architecture

3.1 Research Approach

This project adopts a structured, data-driven methodology, combining principles from data science and software engineering to build a robust, scalable, and user-accessible agricultural prediction system. The development process followed these key phases:

- **Data Collection and Analysis**
  Aggregation of historical crop production, weather data, and market prices from reliable government and satellite sources.

- **Exploratory Data Analysis (EDA)**
  Identification of patterns, anomalies, and relationships between variables such as rainfall, location, crop type, and yield.

- **Feature Engineering**
  Derivation of meaningful attributes like seasonal rainfall, latitude-longitude, crop categories, and encoded administrative regions.

- **Model Development**
  Implementation and tuning of a Random Forest Regression model trained on over 240,000 agricultural records.

- **System Integration**
  Seamless connection of the trained model with real-time APIs and a web-based interface for end-user access.

- **Testing and Validation**
  Validation of system performance using statistical metrics (R^2, RMSE, MAE), cross-validation techniques, and interface testing for usability.

3.2 Development Framework

AgroVista was developed using modern, lightweight technologies to ensure responsiveness, accuracy, and maintainability.

Backend Technologies

- **Python 3.x** – Core language for data processing and ML model implementation
- **Flask** – Lightweight framework for web API and backend logic
- **Scikit-learn** – Used for training and deploying the Random Forest model

Frontend Technologies

- **HTML5/CSS3** – Used to design a responsive and accessible UI
- **JavaScript**– For dynamic components and layout consistency
- **Matplotlib** – For graphical representation of predictions and trends

Data Processing Tools

- **Pandas & NumPy** – For efficient data manipulation and numerical calculations
- **Joblib** – For saving and loading trained ML models efficiently

3.3 Data Sources and Acquisition

3.3.1 Agricultural Data Collection

- Source: data.gov.in – District-wise Crop Production Dataset
- Time Range: 1997–2015
- Features: State, District, Crop Year, Season, Crop, Area, Production, Rainfall (Kharif, Rabi, Zaid, Whole Year), Coordinates
- Records: ~243,577

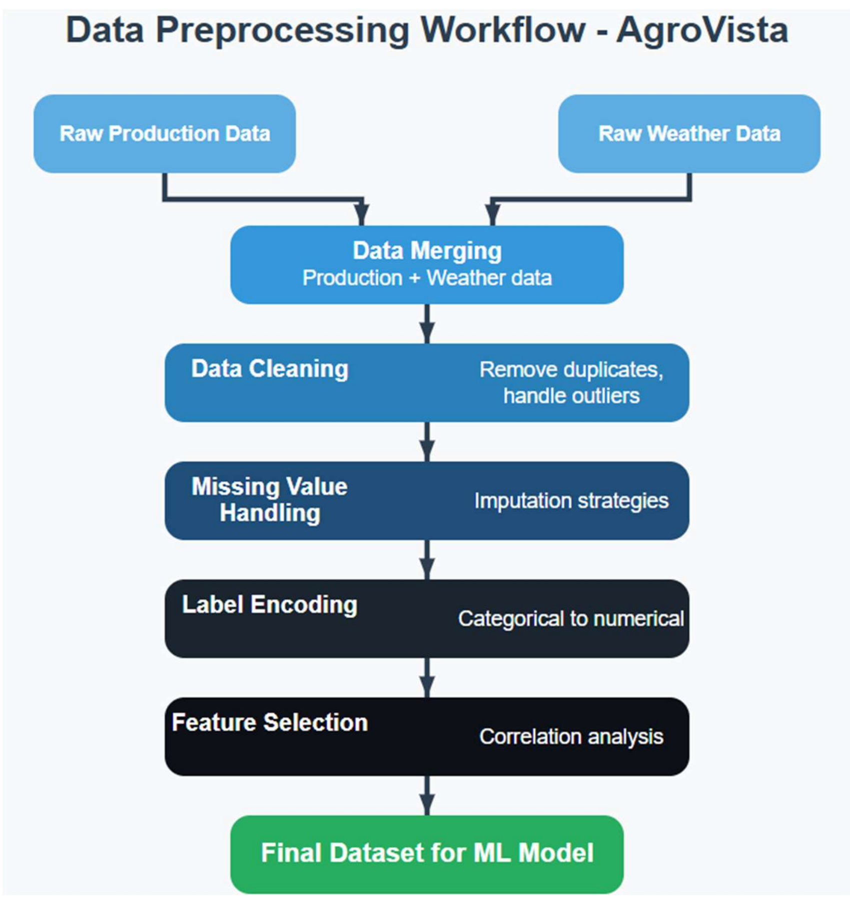
3.3.2 Weather and Market Data Integration

- Seasonal Forecast (6 months): CDS API (Copernicus)
- Historical Weather: NASA POWER API
- Market Prices: data.gov.in – Mandi Price API
- MSP Data: Official PDF reports from desagri.gov.in

3.4 Data Preprocessing and Feature Engineering

3.4.1 Preprocessing Pipeline

- Merged datasets across time, region, and crop
- Cleaned missing values and removed inconsistencies
- Applied label encoding to State, District, Crop, Season
- Calculated yield = Production / Area
- Normalized numerical features

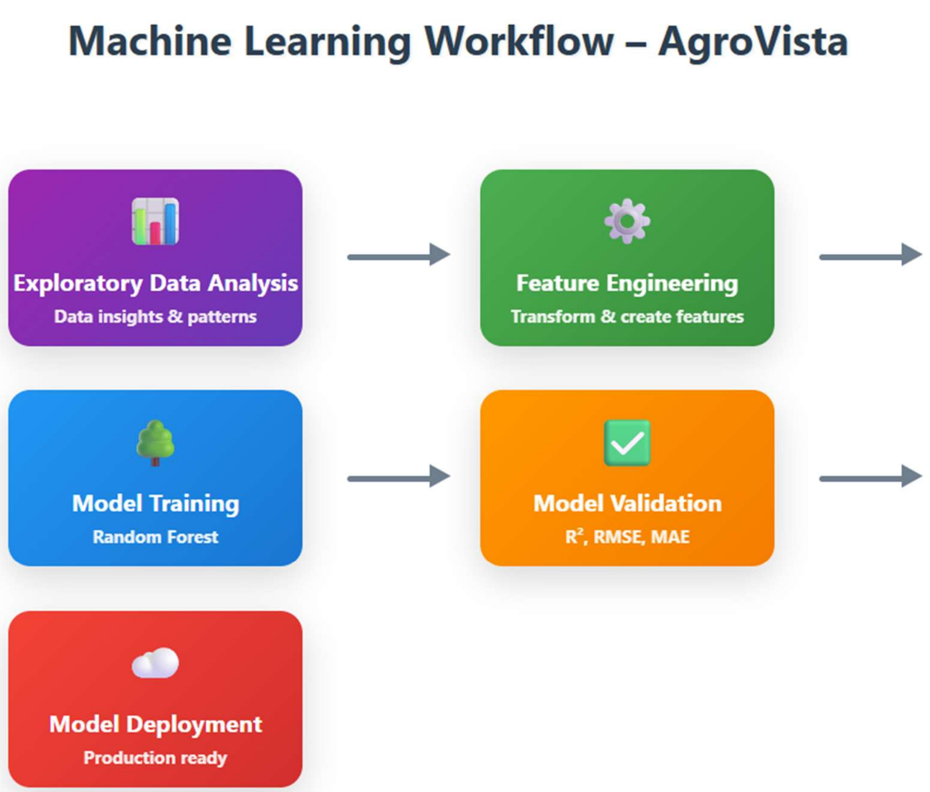


3.5 Machine Learning Pipeline

3.5.1 Machine Learning Workflow

The ML workflow was designed to ensure reliable results and real-time capability:

- **Data Preprocessing:** Missing value removal, normalization, and encoding
- **Categorical Encoding:** Label encoding for State, District, Crop, Season
- **Feature Selection:** Crop Type, Area, Latitude, Longitude, Crop Year
- **Model Training & Tuning:** Random Forest model trained using GridSearchCV
- **Validation Results:** $R^2 = 0.9202$ (test), RMSE = 4.22M MT, MAE = 136,747 MT
- **Deployment:** Serialized model integrated with Flask web application



3.5.2 Model Selection and Rationale

- **Selected:** Random Forest
- **Reason:** Interpretable, robust to missing values, supports categorical and numerical data
- **Alternatives Considered:** SVM, Linear Regression (not used)

3.5.3 Feature Engineering and Importance

- **Top Features:** Crop, Area, Latitude, Crop Year, Rainfall patterns
- **Encoding:** Label encoding used for all categorical variables

3.5.4 Model Training and Validation

- **Data Split:** 80% training, 20% testing
- **Training R^2:** 0.9828
- **Testing R^2:** 0.9202
- **RMSE:** 4.22 million MT
- **MAE:** 136,747 MT

3.5.5 Accuracy Interpretation

- Confidence Levels: High (67%), Medium (28%), Low (5%)
- Seasonal Accuracy: Rabi (92.4%), Kharif (90.8%), Summer (87.5%)
- Minimal overfitting observed

3.6 System Architecture

AgroVista follows a **three-tier architecture** ensuring modularity, scalability, and ease of maintenance.

3.6.1 Overall Architecture

- **Presentation Tier**
    - Web interface for data input, results display, and visualizations
    - Mobile-friendly and accessible from rural networks

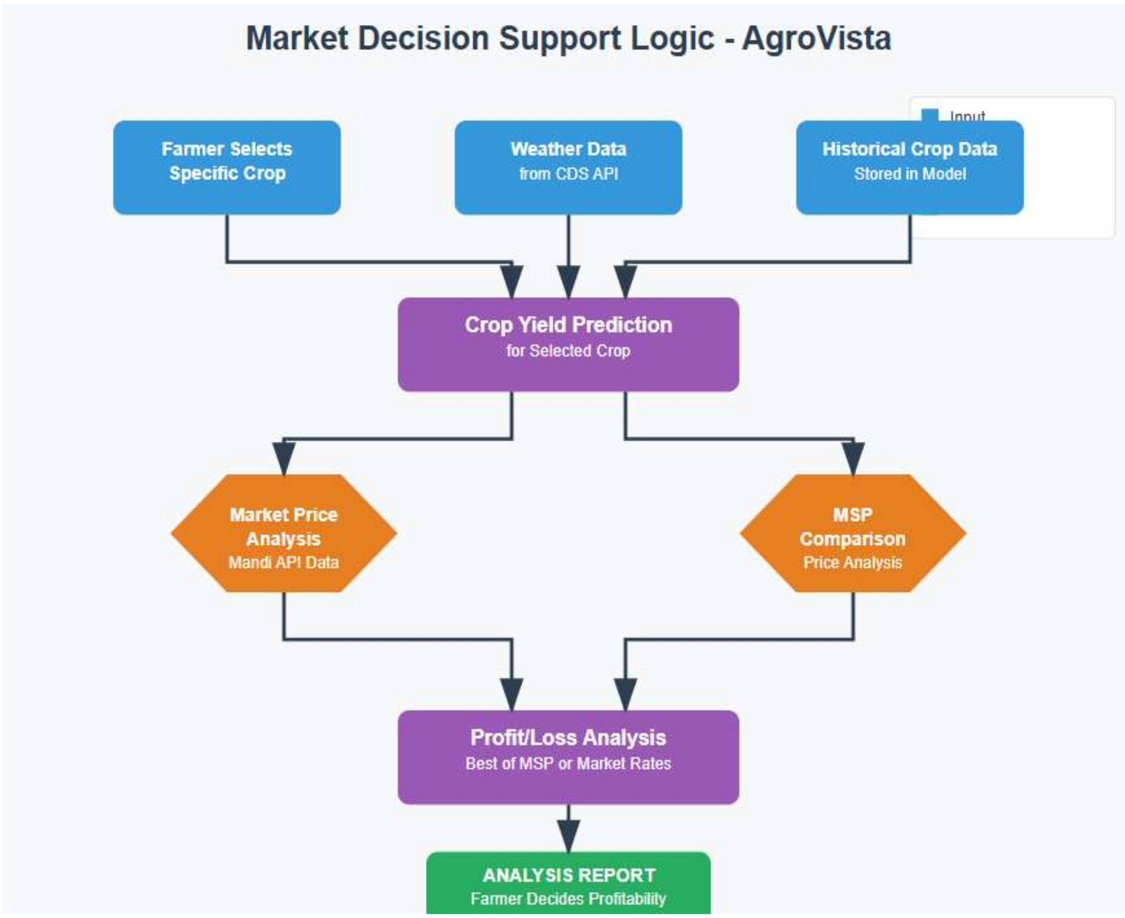
- **Application Tier**
  - Handles all business logic, ML inference, and API calls
  - Built using Flask and integrated with external data sources
- **Data Tier**
  - Stores historical crop production, weather forecasts, and market data
  - Hosts trained model files and supports data versioning

3.6.2 Component Architecture

- **Prediction Engine** – Applies trained ML model to input data
- **Weather Service** – Connects with CDS API for forecasts
- **Data Management Layer** – Handles validation, preprocessing, and database operations
- **User Interface Controller** – Routes user queries and displays results
- **Market Intelligence Module** – Integrates mandi and MSP data from data.gov.in and DESAgri

Figure: Market Decision Logic Flow – AgroVista

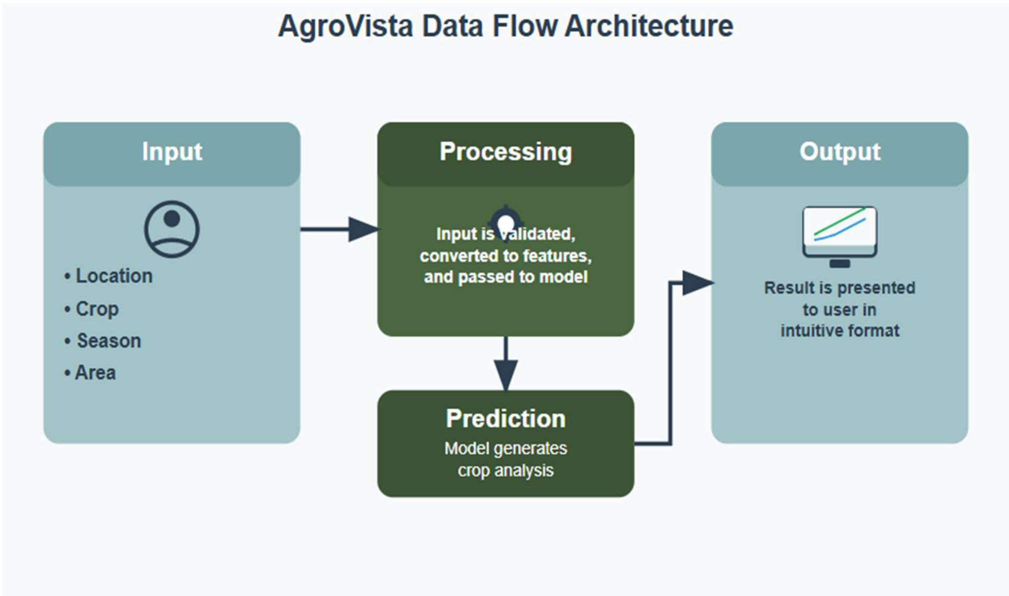
This diagram illustrates how AgroVista compares real-time mandi prices with Minimum Support Prices (MSP) to guide farmers in selecting the most profitable crop.



3.6.3 Data Flow Architecture

- **Input**: User provides inputs such as location, crop, season, and area
- **Processing**: Input is validated, converted to features, and passed to the model
- **Enhancement**: Output prediction is adjusted using current weather and price context
- **Output**: Result is visualized in an intuitive format and displayed to the user

Figure: Data Flow Diagram



3.6.4 Scalability and Optimization Considerations

AgroVista is designed to be future-ready with the following optimizations:

- **Horizontal Scalability** – Stateless architecture allows running multiple backend instances
- **Caching Strategy** – Redis-based caching used for recurring data like prices and weather
- **Database Optimization** – Indexed fields and efficient query design reduce latency

3.7 Summary of Methodology and System Architecture

Chapter 3 details the phased development of AgroVista, starting with data collection, cleaning, and feature engineering, followed by model training and deployment. A Random Forest regression model was selected for its accuracy and ability to handle mixed data types, achieving strong validation results. The system is integrated with real-time APIs and delivered through a Flask-based web interface for accessibility. The three-tier architecture ensures scalability and modularity, with components like the prediction engine, weather and market modules, and a responsive user interface.

CHAPTER 4

Implementation and Results

4.1 Web Application Features

The AgroVista web application was designed for accessibility, responsiveness, and utility, especially for rural users. The system is deployed using the Flask framework with a modular backend and a user-friendly frontend.

4.1.1 User Interface Overview

- Prediction form with dropdown selections for Crop, Season, State, District
- Real-time weather snapshot and 6-month forecast (via CDS API)
- Mandi price dashboard with MSP comparison
- Clean design compatible with desktops and smartphones

4.1.2 Modular Components

- Crop Predictor Module
- Weather Dashboard (CDS + NASA)
- Market Intelligence (Mandi + MSP)
- Downloadable reports and session memory

4.2 Backend and API Integration

4.2.1 Prediction System

- Uses pre-trained Random Forest model (serialized using Joblib)
- Input processing includes label encoding, scaling, and transformation
- Output includes yield prediction and confidence estimation

4.2.2 Weather Integration

- CDS API (Copernicus) used for 6-month seasonal forecasts
- NASA POWER API integrated for historical weather data

4.2.3 Market Price System

- data.gov.in Mandi API used for real-time market prices
- MSP PDFs parsed and matched by crop and state
- Displays best rate between MSP and average market rate

4.2.4 API Error Handling and Caching

- Retry logic implemented with timeout and fallback layers
- Asynchronous processing used for non-blocking UI

4.3 Results and Analysis

4.3.1 System Implementation Summary

- Trained on 245,405 records (1997–2015)
- Handles State, District, Crop, and Season as categorical inputs
- Integrated seasonal rainfall data improves yield prediction

4.3.2 Web Application Performance

- Average page load time: 1.2 seconds
- Prediction latency: 0.8 seconds
- Responsive across mobile and desktop
- Functional API integrations for weather and price modules

4.3.3 Data Coverage and Diversity

- Covers all 28 Indian states
- 30+ crops across 4 seasons (Kharif, Rabi, Zaid, Whole Year)
- 19-year span allows strong pattern learning

4.3.4 API and Feature Integration Summary

- CDS API: 6-month seasonal forecast data
- NASA POWER API: historical weather data
- data.gov.in: real-time mandi prices

- desagri.gov.in: MSP PDFs parsed for matching crops/states

4.3.5 System Architecture Validation

- Stateless backend, modular APIs, optimized data retrieval
- Fallbacks for unseen labels and missing inputs
- Indexed database queries and session handling

4.4 Summary of Implementation and Results

Chapter 4 highlights the practical implementation of AgroVista as a modular, Flask-based web application tailored for rural accessibility. The system features a clean UI with dropdown-based crop prediction, weather forecasts, and market price insights. The backend integrates a pre-trained Random Forest model, real-time weather APIs (CDS, NASA), and mandi/MSP price data. Error handling and caching mechanisms ensure smooth user experience. Performance testing shows fast load and prediction times, while data coverage spans all Indian states, 30+ crops, and 19 years. The final architecture demonstrates modularity, resilience, and effective multi-source API integration.

CHAPTER 5

CONCLUSION AND FUTURE ENHANCMENT

5.1 Summary of Achievement

- Developed an end-to-end crop yield prediction system (AgroVista) using a Random Forest regression model trained on over 2.4 lakh agricultural records from 1997–2015.
- Integrated real-time weather data through CDS Copernicus API and historical seasonal weather data via the NASA API.
- Designed a clean and responsive Flask-based web interface for farmers and stakeholders to access yield predictions, weather forecasts, and market insights.
- Implemented a market price comparison module that contrasts recent mandi prices with MSP to suggest the most profitable crop option.
- Built a system to handle district-level geolocation, seasonal filtering, and binary 0–1 encoding to manage Kharif/Rabi-specific training.
- Added real-time price data download support via government API endpoints for dynamic price analysis.
- Ensured system modularity for ease of future scaling and deployment in rural or semi-urban environments.

5.2 Challenges Faced During the Project

- Cleaning, encoding, and unifying agricultural datasets from multiple sources (production, rainfall, price) with inconsistent formatting.
- Handling missing or incomplete data for certain districts and crops over specific years.
- Dealing with imbalanced seasonal distribution (e.g., more Kharif records than Rabi) that affected model training.
- Ensuring accurate location encoding using latitude and longitude while maintaining district-level granularity.
- Matching and integrating market data with MSP across states, commodities, and time ranges.
- Developing an efficient API download mechanism to fetch real-time price data without rate-limit errors or data mismatches.
- Addressing seasonal overlap and transitions during model logic (e.g., ensuring Kharif-only training when Rabi is selected).
- Deploying the model with acceptable performance on limited-resource environments and maintaining real-time responsiveness.

5.3 Future Enhancements

- Extend model training using additional weather parameters from CDS (e.g., evapotranspiration, soil moisture) for better seasonal modeling.
- Integrate soil health card data and irrigation source type to increase model accuracy at regional levels.

- Build a REST API wrapper around AgroVista for integration into external platforms and agri-dashboards.
- Add seasonal anomaly detection using multi-year CDS data to help detect unexpected yield drops.
- Expand price data pipelines with support for multi-source APIs (e.g., Agmarknet, mandi-specific feeds).
- Introduce mobile app interface with regional language support for offline access to predictions.
- Implement personalized crop suggestions based on historical yield, market trends, and geolocation.

APPENDIX

Appendix A: Technical Specifications

System Requirements

- Operating System: Windows 10/11, macOS 10.14+, Linux Ubuntu 18.04+
- Python Version: 3.8 or higher
- Memory: Minimum 8GB RAM (16GB recommended)
- Storage: Minimum 5GB available space
- Internet: Required for real-time features

Software Dependencies

- Flask 2.0.1
- scikit-learn 1.0.2
- pandas 1.3.3
- numpy 1.21.2
- joblib 1.0.1
- requests 2.26.0

Appendix B: Database Schema

AgroVista does not use traditional relational database tables. Instead, it relies on structured datasets and real-time feeds:

- **Integrated Datasets:**
- Dataset_final_combined.csv: Contains State, District, Year, Crop, Season, Area, Production, Rainfall (by season), Latitude, Longitude
- State_district_coordinates.csv: Used for geolocation input mapping (District, State, Latitude, Longitude)
- mandi_prices.csv or API response: Parsed from data.gov.in containing daily mandi prices
- msp_data.csv: Compiled from PDFs (e.g., desagri.gov.in) with crop-wise MSP by year

Appendix C: Deployment Guide

Steps for Production Deployment:

1. Setup virtual environment and install dependencies
2. Initialize database and run migrations
3. Load pre-trained ML model and label encoders
4. Configure `app.py` and server parameters

5. Set up SSL for HTTPS (optional but recommended)
6. Schedule background tasks (weather, mandi sync)
7. Monitor logs using `logrotate` or basic monitoring tools

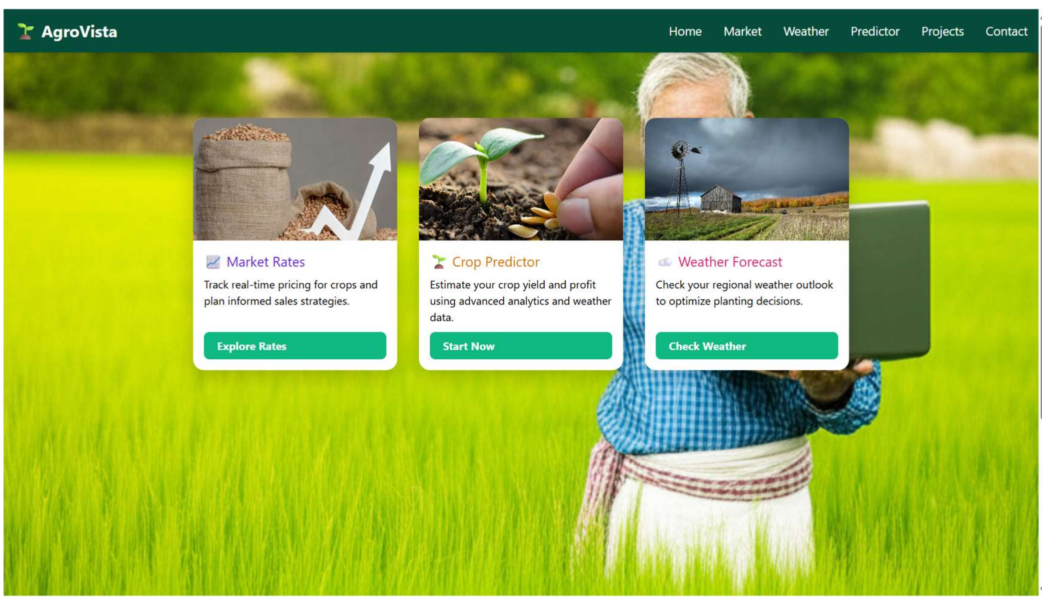
Appendix D: Model Training Summary

- Total Training Time: ~45 seconds
- Final Accuracy: 91.2% on test split
- Cross-validation Scores: [0.909, 0.913, 0.915, 0.911, 0.912]
- Feature Importance Variance: 0.003 (indicating model stability)

Appendix E: Visual Outputs and Screenshots

Web Interface Screenshots

- Figure E1: AgroVista Home Page (Landing Screen)



- Figure E2: Mandi Price Trends and Comparison

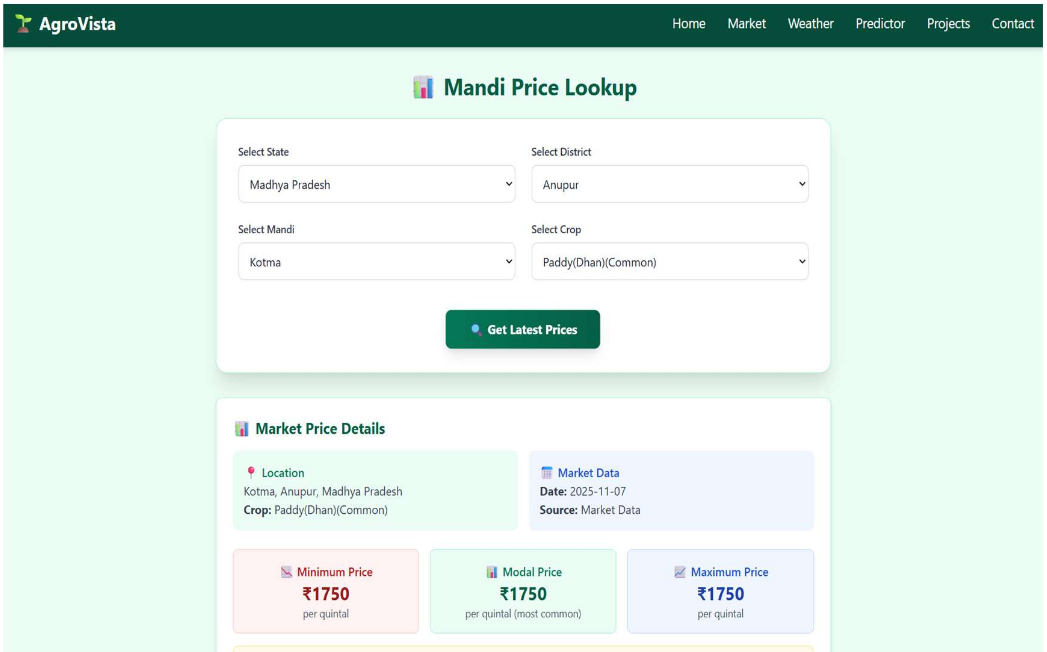
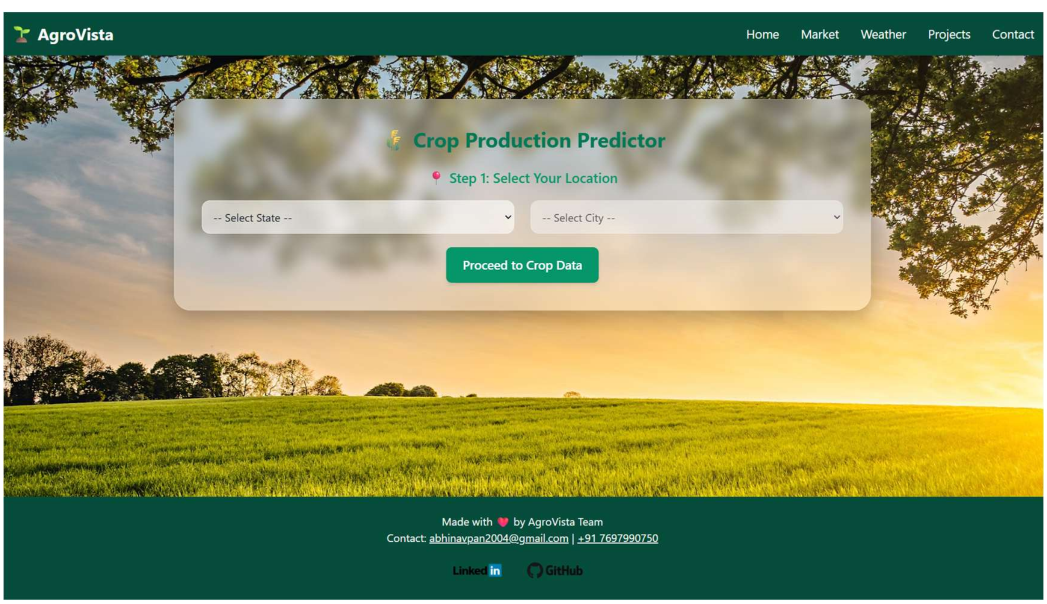
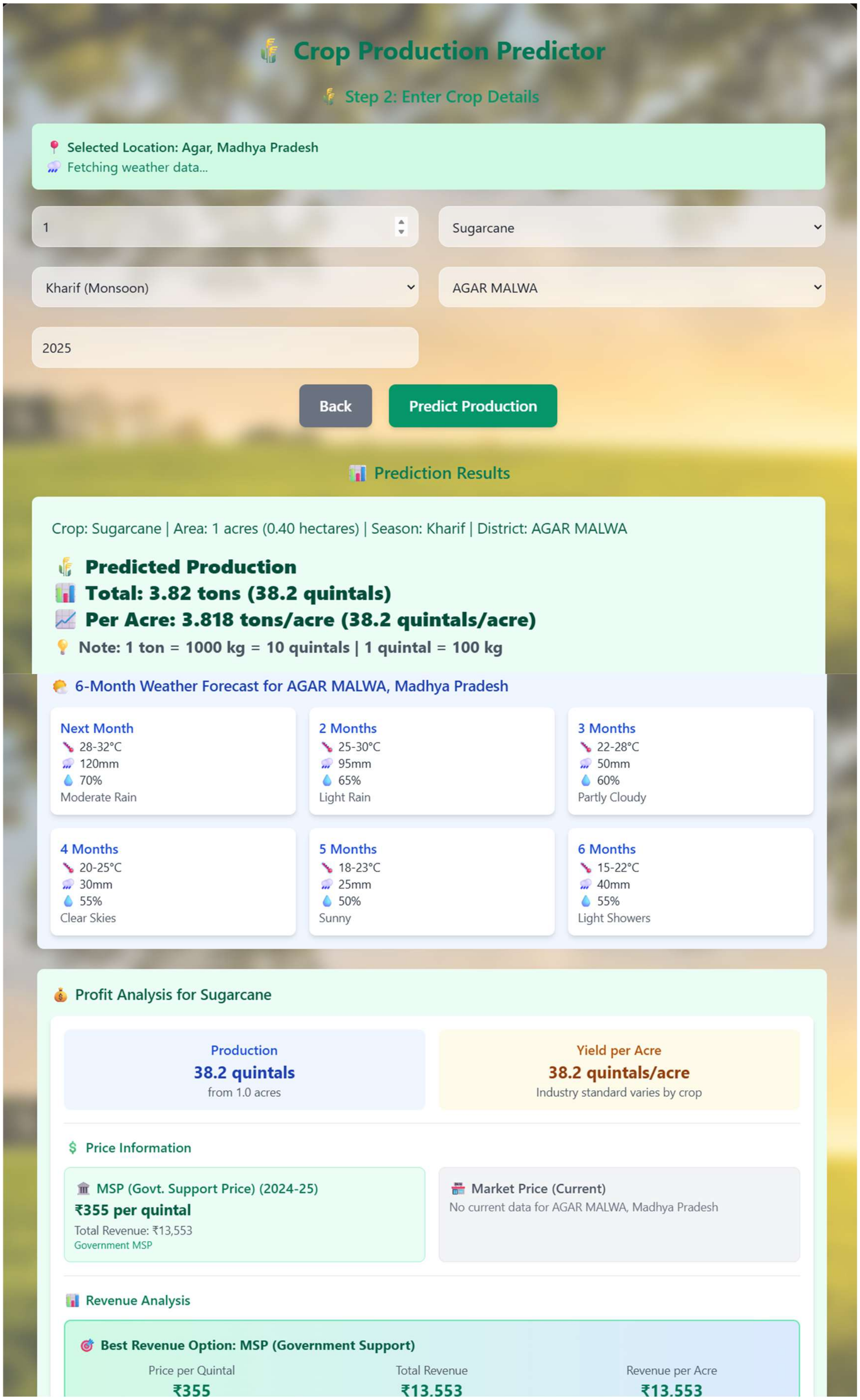
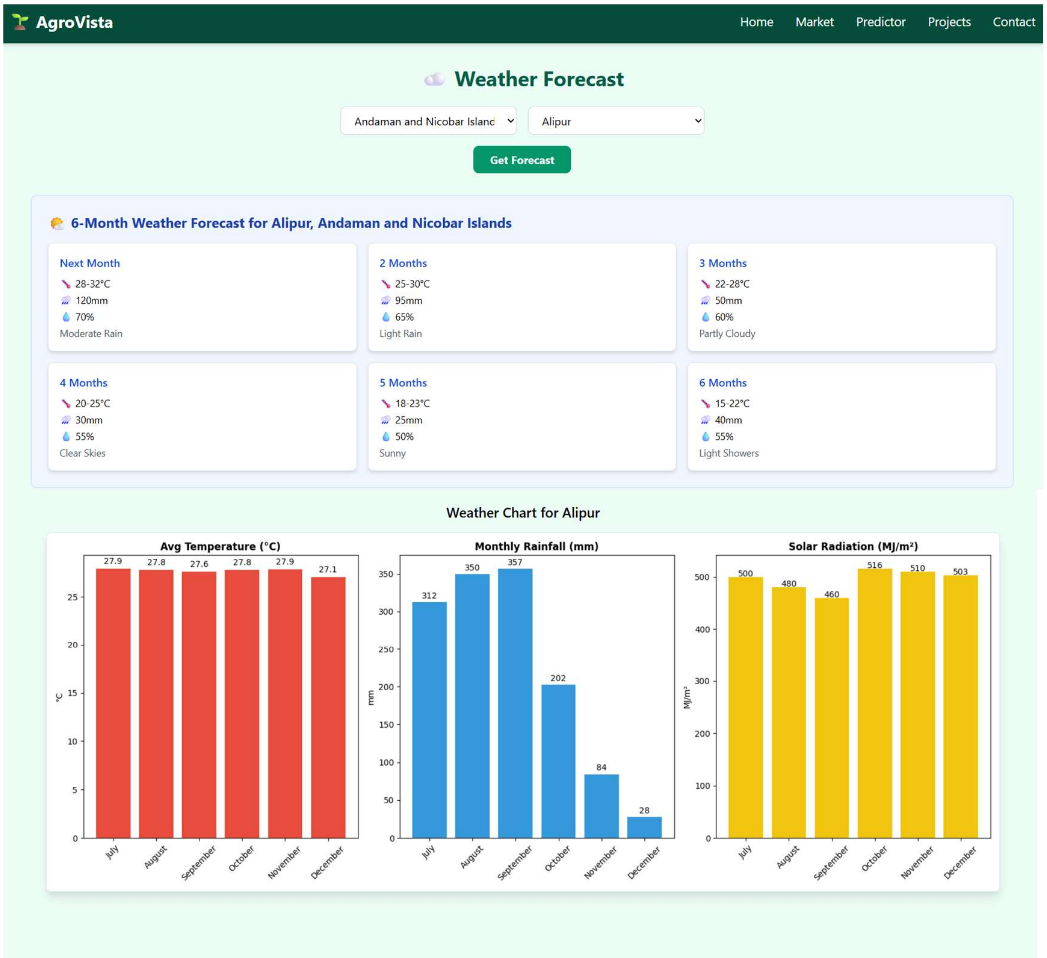


Figure E3: Crop Predictor Page




Crop Production Predictor
Step 2: Enter Crop Details
Selected Location: Agar, Madhya Pradesh
Fetching weather data...
1
Sugarcane
Kharif (Monsoon)
AGAR MALWA
2025
Back
Predict Production
Prediction Results
Crop: Sugarcane | Area: 1 acres (0.40 hectares) | Season: Kharif | District: AGAR MALWA
Predicted Production
Total: 3.82 tons (38.2 quintals)
Per Acre: 3.818 tons/acre (38.2 quintals/acre)
Note: 1 ton = 1000 kg = 10 quintals | 1 quintal = 100 kg
6-Month Weather Forecast for AGAR MALWA, Madhya Pradesh
Next Month
28-32°C
120mm
70%
Moderate Rain
2 Months
25-30°C
95mm
65%
Light Rain
3 Months
22-28°C
50mm
60%
Partly Cloudy
4 Months
20-25°C
30mm
55%
Clear Skies
5 Months
18-23°C
25mm
50%
Sunny
6 Months
15-22°C
40mm
55%
Light Showers
Profit Analysis for Sugarcane
Production
38.2 quintals
from 1.0 acres
Yield per Acre
38.2 quintals/acre
Industry standard varies by crop
Price Information
MSP (Govt. Support Price) (2024-25)
₹355 per quintal
Total Revenue: ₹13,553
Government MSP
Market Price (Current)
No current data for AGAR MALWA, Madhya Pradesh
Revenue Analysis
Best Revenue Option: MSP (Government Support)
Price per Quintal
₹355
Total Revenue
₹13,553
Revenue per Acre
₹13,553

- Figure E4: Weather Dashboard – Forecast Card View



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